MANAGEMENT INFORMATION SYSTEMS
ICT2162
LECTURE 05

Ms. Shelomi De Alwis
Department of ICT
Faculty of Technology
University of Ruhuna

Operational Excellence and Customer Intimacy - Part I

What we discuss today?

- Enterprise Systems(ERP)
  - Characteristics
  - How it works
  - Business Value
- Supply Chain Management Systems
  - Up Stream and Down Stream
  - Information and SCM
  - Bullwhip effect
  - How IS facilitate SCM
  - Supply chain management software
  - Global supply chain issues
  - Supply chain management systems
  - Business value of SCM systems
  - The emerging internet-driven supply chain

Objectives

- Understand how do enterprise systems help businesses achieve operational excellence?

- Understand how do supply chain management systems coordinate planning, production, and logistics with suppliers?

AVON

the company for women



Enterprise Systems

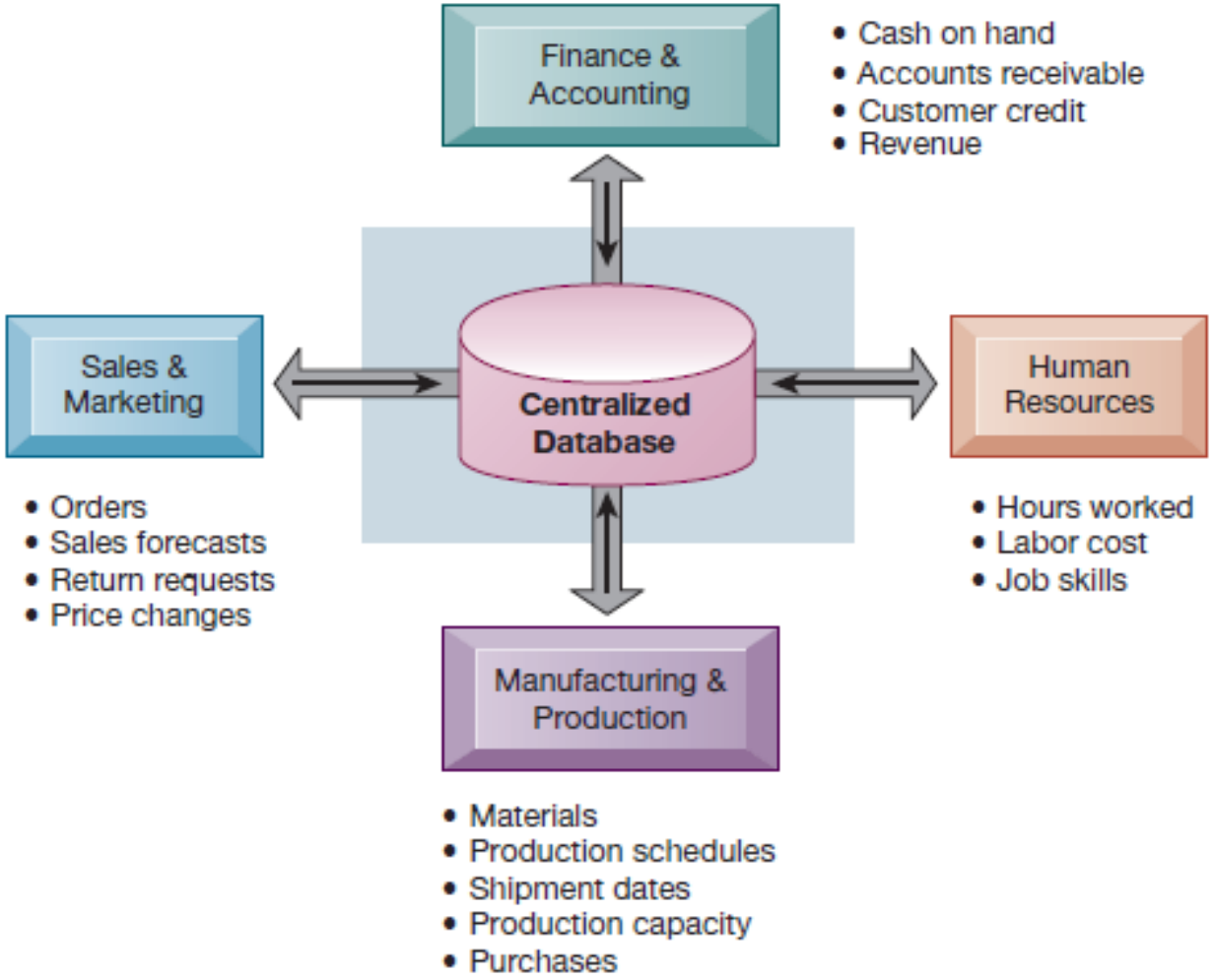
- Also called “Enterprise Resource Planning (ERP) systems”
- Suite of integrated software modules and a common central database
- Collects data from many divisions of firm for use in nearly all of firm’s internal business activities
- Information entered in one process is immediately available for other processes

Enterprise Systems

Characteristics of an Enterprise System

1. Real Time Operations
2. Integration of all Business Activities
3. Flow of Information

Enterprise Systems – How it Works



Enterprise Systems

- Enterprise Software
  - Built around thousands of predefined business process that reflect best practices
- Business processes supported by enterprise systems
  - Finance/accounting: General ledger, accounts payable, etc.
  - Human resources: Personnel administration, payroll, etc.
  - Manufacturing/production: Purchasing, shipping, etc.
  - Sales/marketing: Order processing, billing, sales planning, etc.
- To implement Enterprise Software, firms:
  - Select functions of system they wish to use
  - Map business processes to software processes
    - ✓ Use software's configuration tables for customizing

Enterprise Systems

- Business value of enterprise systems
  - Increase operational efficiency
  - Provide firm wide information to support decision making
  - Enable rapid responses to customer requests for information or products
  - Include analytical tools to evaluate overall organizational performance

Supply Chain Management Systems

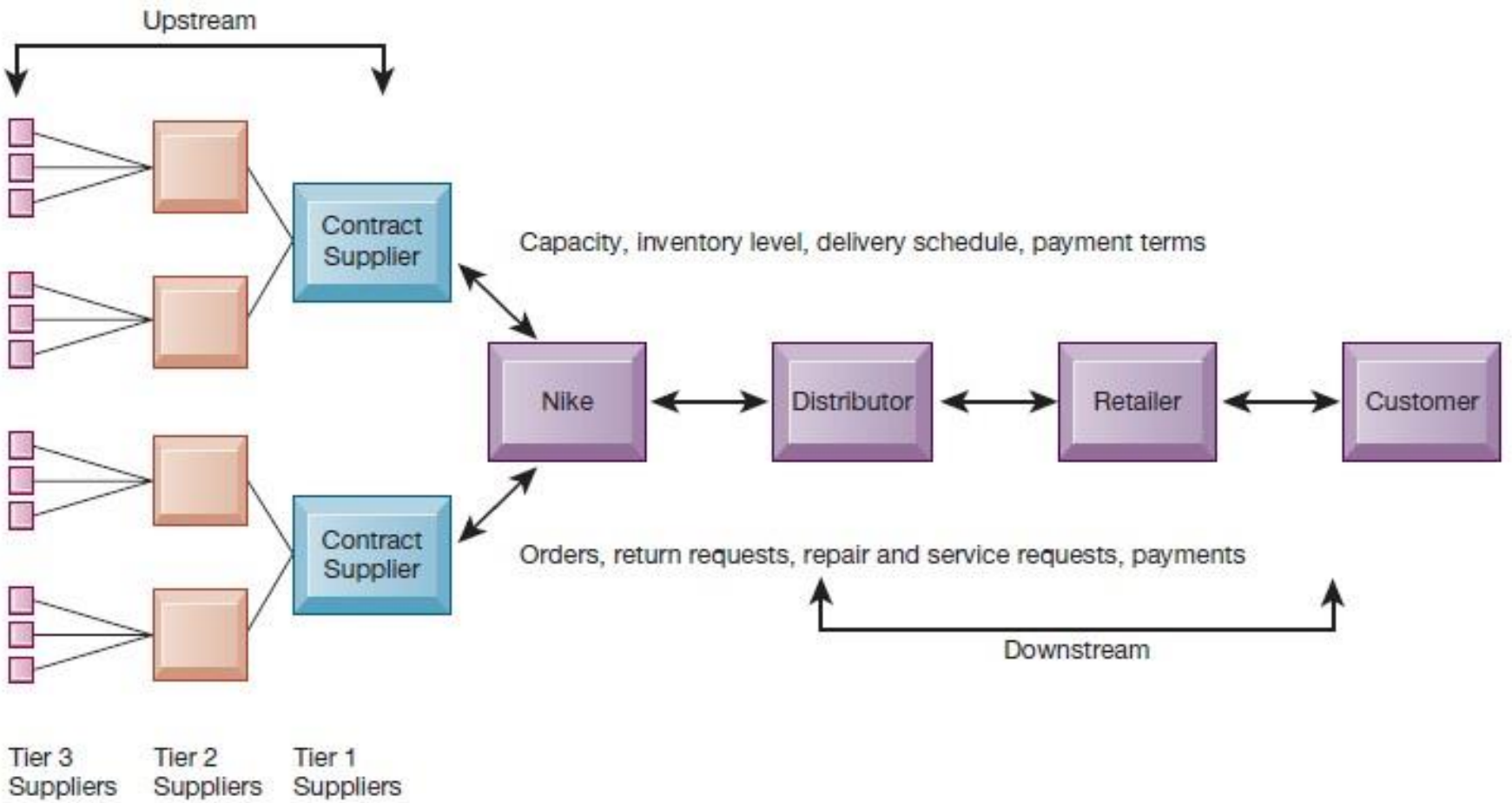


Supply Chain Management Systems

- Supply chain:
  - Network of organizations and business processes for:

    ✓ Procuring raw materials
    ✓ Transforming them into intermediate and finished products
    ✓ Distributing the products

Supply Chain Management Systems Example



Nike's Supply Chain

Supply Chain Management Systems

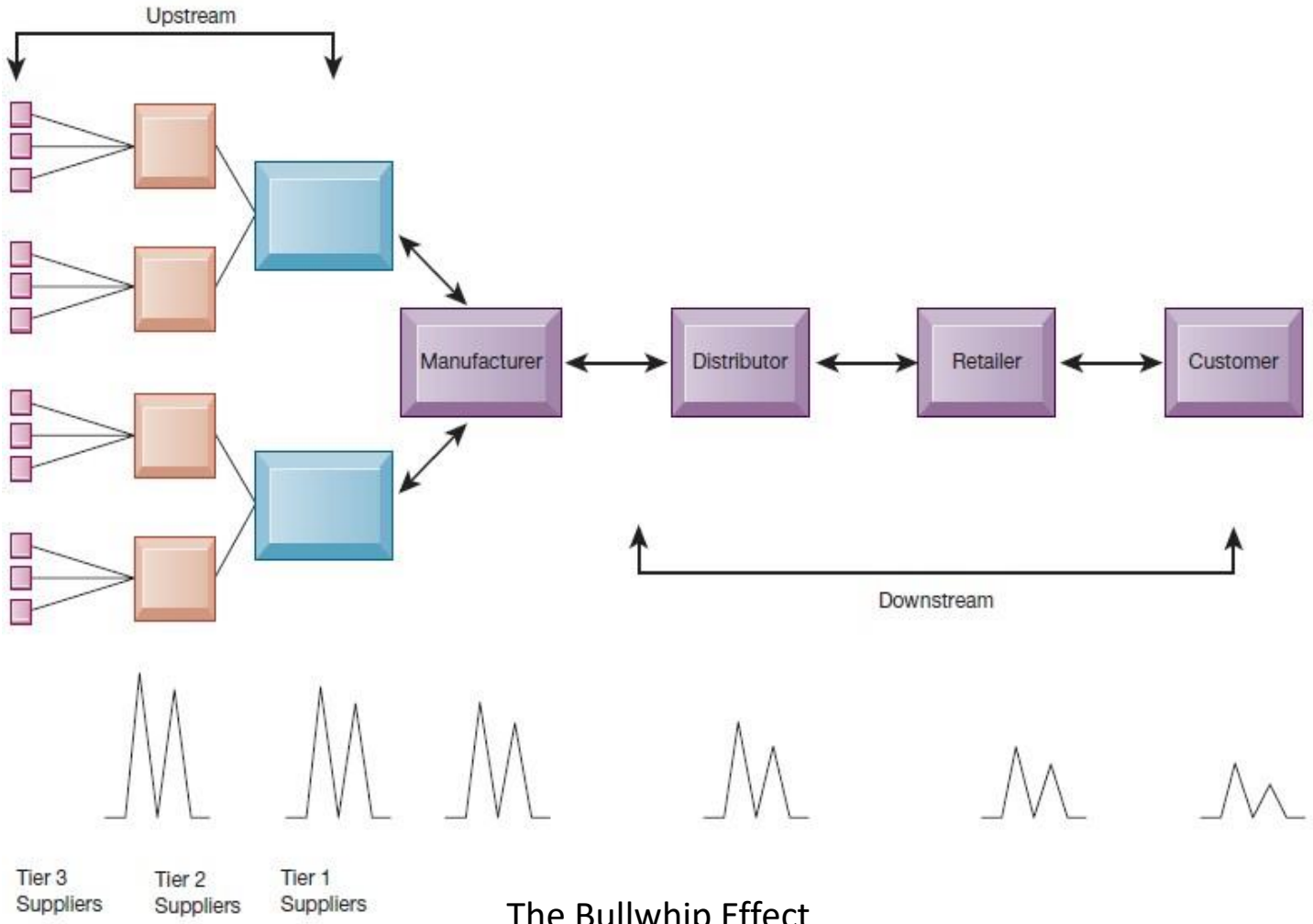
- Upstream supply chain:
  - Firm's suppliers, suppliers' suppliers, processes for managing relationships with them

- Downstream supply chain:
  - Organizations and processes responsible for distributing and delivering products to customers

Supply Chain Management Systems

- Information and supply chain management
  - Supply chain inefficiencies waste as much as 25 percent of a company's operating costs.
  - Just-in-time strategy:
    - Components arrive as they are needed
    - Finished goods shipped after leaving assembly line
  - Safety stock
    - Acts as a buffer for the lack of flexibility in the supply chain.
  - Bullwhip effect
    - Information about product demand gets distorted as it passes from one entity to next across supply chain

Supply Chain Management Systems



The Bullwhip Effect

Supply Chain Management Systems

- How information systems facilitate supply chain management
  - Decide when and what to produce, store, and move
  - Rapidly communicate orders
  - Track the status of orders
  - Check inventory availability and monitor inventory levels
  - Reduce inventory, transportation, and warehousing costs
  - Track shipments
  - Plan production based on actual customer demand
  - Rapidly communicate changes in product design

Supply Chain Management Systems

Supply chain management software

- Supply chain planning systems
  - Model existing supply chain
  - Optimize sourcing, manufacturing plans
  - Establish inventory levels(Raw, Intermediate, finished)
  - Where to store finished products
  - Identifying transportation modes
  - Facilitate Demand planning
- Supply chain execution systems
  - Manage flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner

Supply Chain Management Systems

Global Supply Chains

- Internet helps companies manage many aspects of global supply chains
  - Sourcing, transportation, communications, international finance etc.

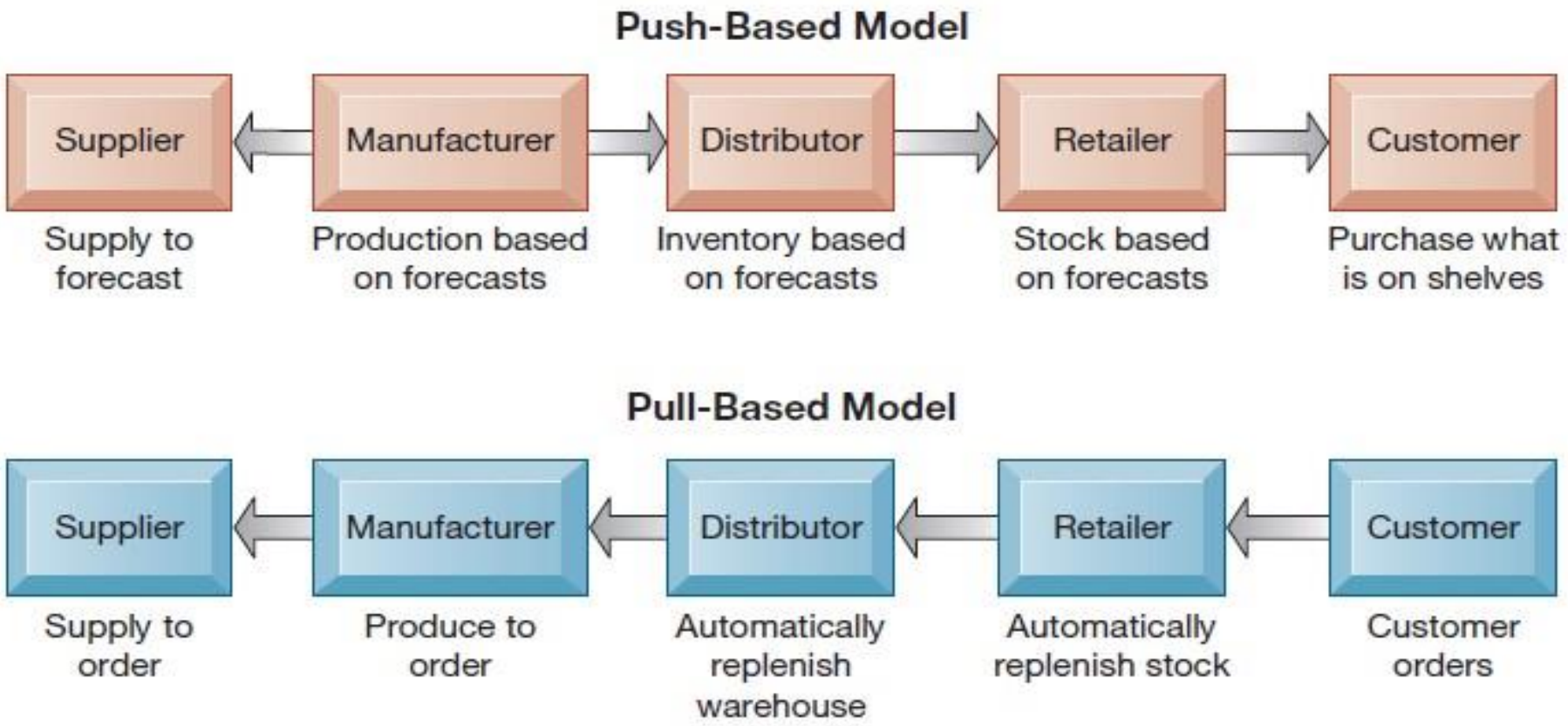
- Global supply chain issues
  - Global supply chains typically span greater geographic distances and time differences
  - More complex pricing issues (local taxes, transportation, etc.)
  - Foreign government regulations
  - Cultural differences

Supply Chain Management Systems

Supply chain management systems

1. Push-based model (build-to-stock)
   - Schedules based on best guesses of demand
2. Pull-based model (demand-driven)
   - Customer orders trigger events in supply chain
3. Sequential supply chains
   - Information and materials flow sequentially from company to company
4. Concurrent supply chains
   - Information flows in many directions simultaneously among members of a supply chain network

Supply Chain Management Systems



Supply Chain Management Systems

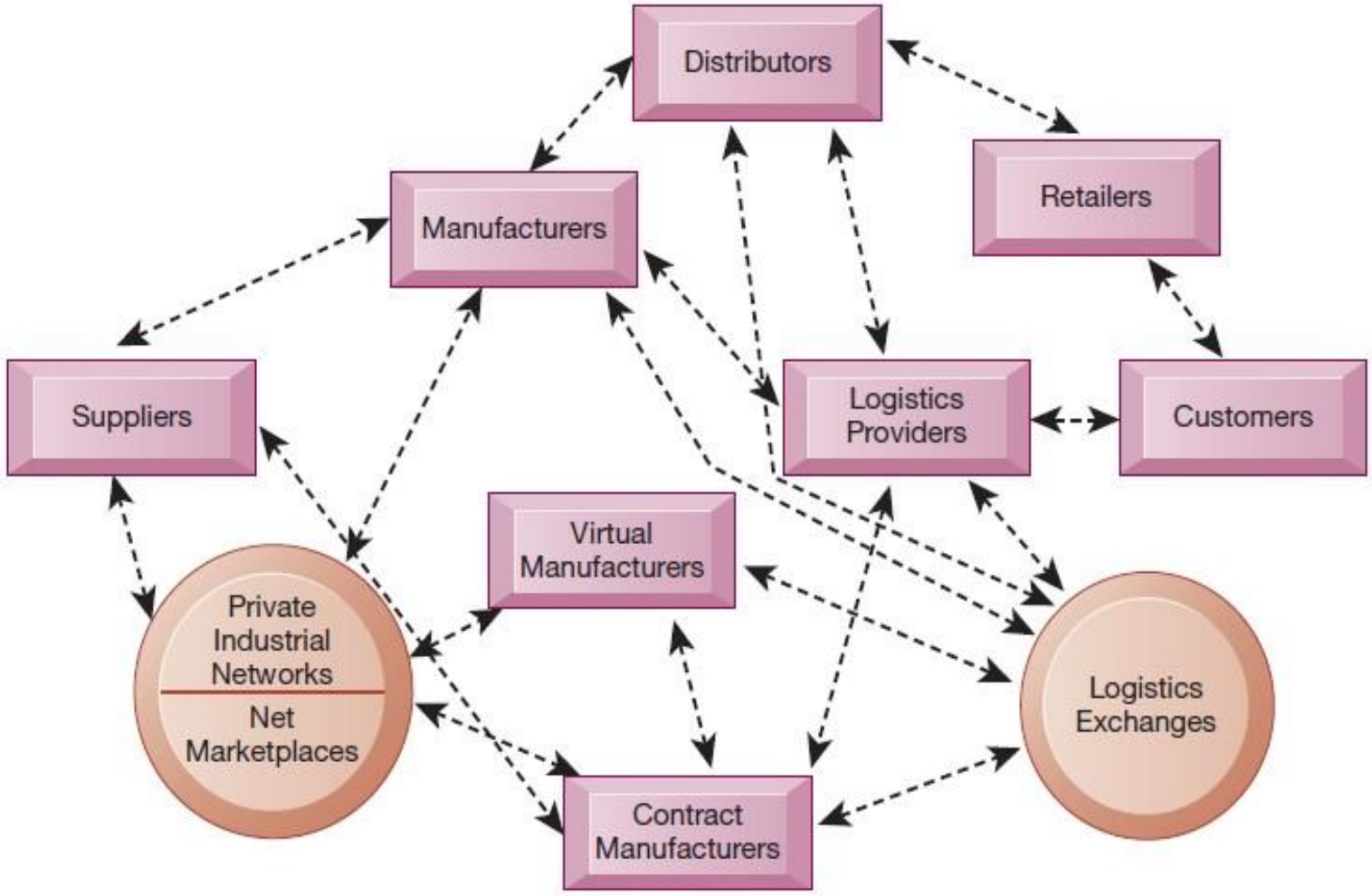
Business value of SCM systems

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
- Reduced supply chain costs lead to increased profitability
- Increased sales

Supply Chain Management Systems

The emerging internet driven supply chain

The emerging Internet-driven supply chain operates like a digital logistics nervous system. It provides multidirectional communication among firms, networks of firms, and e-marketplaces so that entire net works of supply chain partners can immediately adjust inventories, orders, and capacities.



Summary

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  - Characteristics
  - How it works
  - Business Value
- Supply Chain Management Systems
  - Up Stream and Down Stream
  - Information and SCM
  - Bullwhip effect
  - How IS facilitate SCM
  - Supply chain management software
  - Global supply chain issues
  - Supply chain management systems
    - Push, pull, sequential, concurrent
  - Business value of SCM systems
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Questions ???



Thank you…